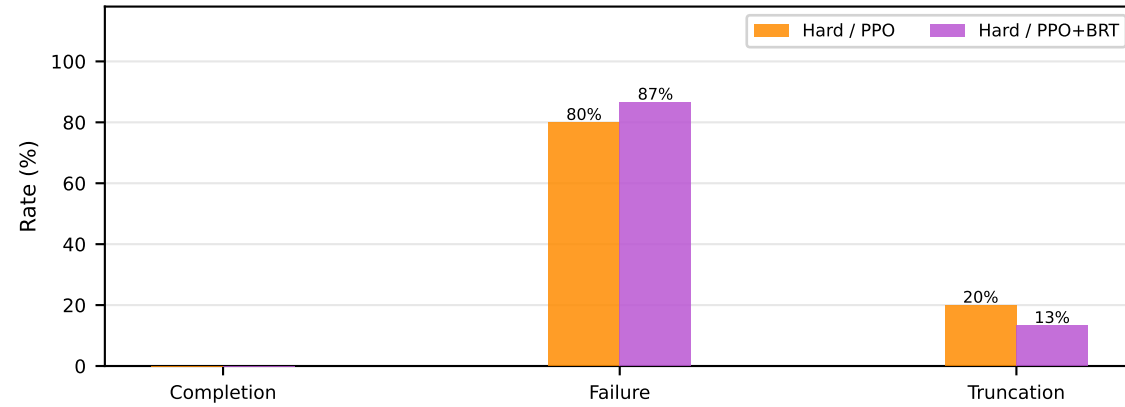
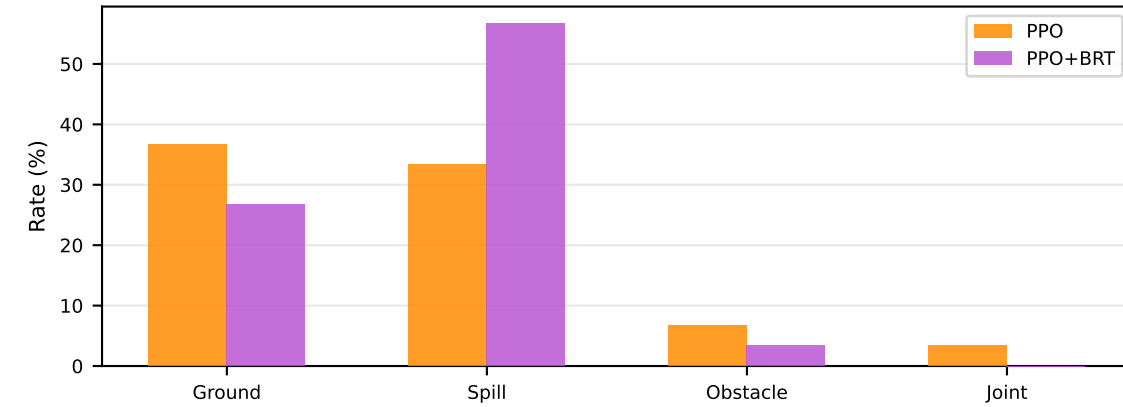


Safety evaluation: PPO vs PPO + BRT safety filter
PPO hard-start failure 80% | PPO+BRT hard-start failure 87%

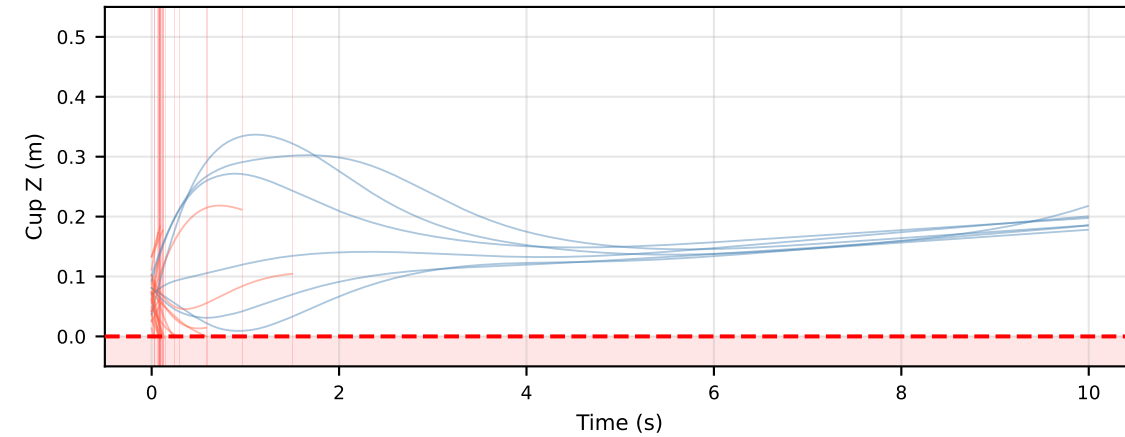
(A) Episode outcomes



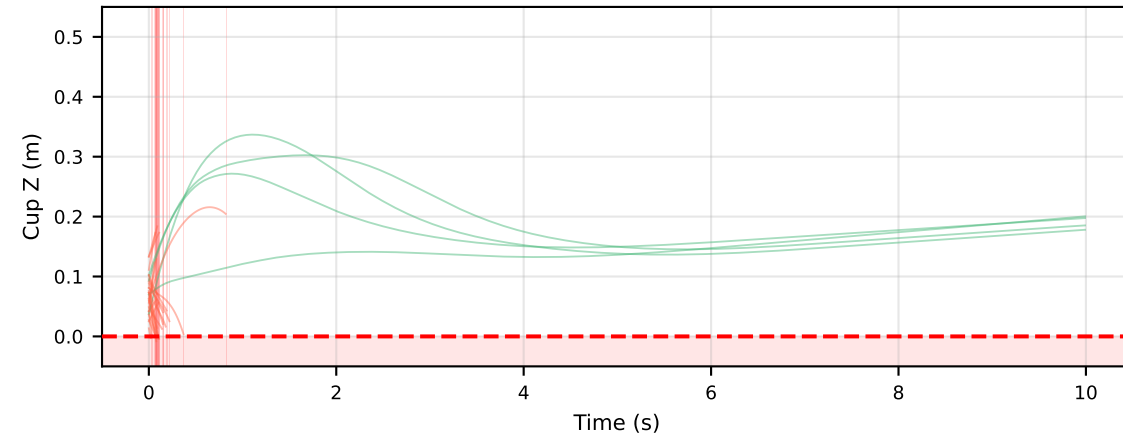
(B) Failure type breakdown (hard starts)



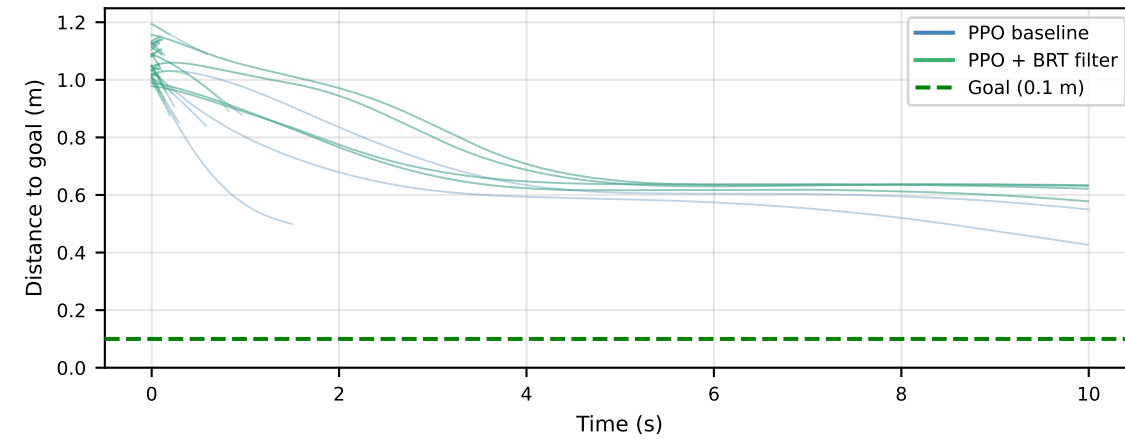
(C) Hard starts — PPO baseline (cup height)



(D) Hard starts — PPO + BRT filter (cup height)



(E) Progress toward goal (hard starts)



(F) BRT filter activity (hard starts)

